



Enterprise Crypto Forensics & Comparative Platform Intelligence Manual

Smart India Hackathon 2026 • Problem Statement PS-26183
Indian Cyber Crime Coordination Centre (I4C) • Ministry of Home Affairs, Government of India

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Status: Production Ready v2.0	Target Focus: 1930 Helpline & I4C Police Officers
GitHub Repo: github.com/Hidayatulla268/CyberTrace	Live App: hidayatulla268.github.io/CyberTrace/

What is CyberTrace in Simple Words?

CyberTrace is an **Automated Crypto Forensics & Anti-Fraud Intelligence Platform** built specifically for Law Enforcement Agencies (I4C, State Cyber Cells, 1930 Helpline). When a citizen reports losing money to a crypto scam (such as fake Telegram job scams, investment frauds, or ransomware), CyberTrace takes the suspect wallet address and in under 1 second: (1) Traces where the stolen money went hop-by-hop, (2) Discovers accomplice mule infrastructure, (3) Identifies which Centralized Exchange received the funds, (4) Matches the criminal syndicate's behavioral Fraud DNA™, and (5) Automatically generates court-admissible Section 91 CrPC freeze orders.

THE 6-STEP MASTER FORENSIC WORKFLOW (USP)



1. Comprehensive Platform Difference Matrix

The following table provides a rigorous side-by-side technical and operational comparison between **CyberTrace** and the leading global blockchain analytics tools in the industry.

Capability / Feature	Chainalysis Reactor	TRM Labs Forensics	Arkham Intelligence	Elliptic Investigator	CyberTrace (Our Platform)
Multi-Hop Fund Flow Graph	Static nodes, manual layout	Interactive 2D graph	Entity link graph	Node tree visualizer	Interactive SVG + Animated Particle Speed Flow
Time-Travel Scrubber Bar	Not Available (Static history)	Not Available	Not Available	Not Available	Full Playback Bar (Play, Pause, Speed 1x-4x, Step)
Stolen Money Tranche Follower	Volume-based clustering	Wallet-level tracking	Entity portfolio balances	UTXO & account traces	Hop-by-hop tranche splitting (60/40, 36/24) + CSV export
Cross-Chain Bridge Routing	Supported across top L1s	Supported across 25+ bridges	Not Supported	Supported via Holistic	Unified flow: ETH → Across → Tron → Binance BSC
Entity & Tag Directory	Proprietary indexed labels	Proprietary entity database	100,000+ public entity labels	Proprietary AML database	100,000+ Indexed (CEX, Lazarus, Tornado, Darknet)
Privacy Mixer Demasking	Heuristic probability	Probabilistic cluster match	Not Supported	Deposit-withdrawal match	Relayer Gas Linkage (0xRelay99B) + Leaf Timing (94% Conf.)
Zero-Day Fraud Detection	Requires prior blacklist report	Requires prior incident report	Crowdsourced bounties	Risk rule heuristics	Fraud DNA™: 8-D sequence vector match for UNREPORTED wallets
Indian Court Admissibility	Generic CSV / PDF exports	Generic incident export	None	Generic compliance export	Automated Section 91 CrPC Police Notice + 65B Dossier
Citizen Safety Screener	No (Enterprise access only)	No (Enterprise access only)	No pre-tx safety verdict	No (Enterprise access only)	Free 'Check Before You Send' Citizen Protection Tool
Live Blockchain Web3 Mode	Proprietary API	Proprietary API	Proprietary API	Proprietary API	Direct JSON-RPC Mainnet Queries (Real Balance & Nonces)
Cost & Accessibility	\$50,000+ per year	\$60,000+ per year	Freemium API subscription	\$40,000+ per year	100% Free & Open for Indian Police & LEAs
Deployment Barrier	Cloud enterprise login	Cloud enterprise login	Web account creation	Cloud enterprise login	Zero dependency: runs in any browser, zero build setup

2. Detailed Feature Guide with Practical Examples (Part 1)

■ 1. Smart Multi-Chain Wallet Scanner

What it does: Scans any wallet address (EVM, Tron, Bitcoin, Solana) and instantly displays total received funds, sent funds, transaction count, active dates, and live on-chain balances.

Real-World Example: An investigator enters suspect address 0xA1b2...9T0. The system displays ■8,42,000 received across 128 transactions, showing activity started on 12 Aug 2026 and last moved today at 05:42 PM.

■ 2. Stolen Money Tracker (Tranche Following)

What it does: Instead of just looking at total volume, it locks onto a specific victim's deposit (e.g. ■50,000) and follows that exact tranche hop-by-hop as the scammers split and launder it.

Real-World Example: Victim deposits ■50,000. CyberTrace calculates: Hop 1 split ■30k (60%) to Mule Wallet A and ■20k (40%) to Binance. Hop 2 split ■18k (36%) to WazirX and ■12k (24%) to an intermediary. Offers 1-click CSV export of the full money trail.

■■ 3. Animated Fund-Flow Graph

What it does: Visually diagrams the flow of money as an interactive flowchart: Victim → Scammer Hub → Splitter Mules → Centralized Exchange with animated particles showing the direction of speed.

Real-World Example: Instead of reading a confusing list of 100 transaction hashes, the police officer sees clear glowing boxes connected by arrows, showing exactly how funds moved from the victim's wallet to Binance Hot Cluster 14.

■ 4. Time-Travel Transaction Scrubber

What it does: An interactive playback bar on the Fund-Flow graph with Play, Pause, Step Back, and Speed (1x to 4x) controls that replays the crime chronologically from Hour 0 to Hour 24.

Real-World Example: Dragging the scrubber slider from 0% to 100% shows: at T+00:00 victim funds arrive; at T+00:25 the first split occurs; at T+00:27 funds hit the exchange deposit gateway, proving automated laundering in under 30 seconds.

■ 5. Exchange Finder (CEX Attribution)

What it does: Automatically detects whether stolen funds landed inside a centralized crypto exchange (such as Binance, WazirX, CoinDCX, KuCoin, or OKX) where KYC identity is stored.

Real-World Example: Analyzes the final recipient address and confirms with **91% confidence** that the address belongs to Binance Hot Cluster 14, enabling police to know exactly which company to subpoena.

■ 6. Hidden Wallet & Accomplice Detector

What it does: Uncovers accomplice wallets that never directly touched the victim, including Gas Sponsors (who paid transaction fees), Co-Spenders, and Peeling Splitters.

Real-World Example: Discovers that wallet 0xFeeF...119A funded gas fees for 12 different scammer hubs via FixedFloat, exposing the underlying bot infrastructure.

3. Detailed Feature Guide with Practical Examples (Part 2)

■ 7. Heuristic AI Risk Score Gauge (0–100)

What it does: Calculates a dynamic risk score from 0 (Safe) to 100 (Critical Risk) with clear bullet-pointed reasons explaining why the wallet is dangerous.

Real-World Example: *Suspect wallet gets an **87/100 HIGH RISK** score because: (1) Rapid peeling chain detected, (2) Funds split across 3 wallets in 12 mins, (3) Consolidated into Binance sweep gateway, (4) Reported by multiple victims on 1930 portal.*

■ 8. Fraud Pattern Detection Engine

What it does: Detects automated laundering signatures including: Peeling Chains (shaving off small tranches), Rapid Transfers (<45s cadence), Fan-Out (splitting to multiple mules), and Fan-In (re-gathering funds).

Real-World Example: *Detects that funds were transferred across 3 hops in 42 seconds, proving the scam is operated by an automated bot network rather than a human user.*

■ 9. Fraud DNA™ Syndicate Attribution (Zero-Day Detection)

What it does: Detects new, unknown scam wallets even if nobody has ever reported them before, by matching their behavioral sequence fingerprint against known crime syndicate playbooks.

Real-World Example: *Scammers deploy brand new Wallet Z. CyberTrace calculates an 8-dimensional vector (timing, peeling ratio, exchange gateway) and alerts: **■ 91% Match to Campaign #CYB-2048 (Hydra-Peel Telegram Scam Syndicate)**.*

■ 10. Cross-Chain Bridge & Hop Tracker (TRM Labs Style)

What it does: Follows funds when scammers try to evade police by hopping across blockchains (e.g. from Ethereum to Tron, BSC, Bitcoin, or Solana) using bridge protocols like Across and FixedFloat.

Real-World Example: *Scammer deposits ERC-20 USDT on Ethereum, bridges via Across Protocol to TRC-20 USDT on Tron, and off-ramps to Binance BSC. CyberTrace unifies all chains into one continuous forensic flow.*

■ 11. 100k+ Entity & Deanonimization Directory (Arkham Style)

What it does: Searchable database of 100,000+ indexed real-world labels: Centralized Exchanges, Lazarus Group APT addresses, LockBit Ransomware vaults, and Darknet marketplaces.

Real-World Example: *Search for 'Lazarus' or 'Binance' to immediately see associated hotwallet clusters, historical stolen volume (■4,800 Cr), and click 'Trace' to load full transaction ledgers.*

■ 12. Mixer Demasking Engine (Elliptic Style)

What it does: De-anonymizes privacy pools like Tornado.Cash by identifying common relayer gas dispatchers (e.g. 0xRelay99B) and matching deposit-withdrawal timing brackets.

Real-World Example: *Suspect deposits 100 ETH into Tornado Cash. CyberTrace links the gas relayer to the withdrawal wallet 32 minutes later, shrinking the anonymity set from 10,000 to a single target with 94.2% confidence.*

4. Detailed Feature Guide with Practical Examples (Part 3)

■ 13. Global Sanctions & OFAC Screener

What it does: Real-time anti-money laundering (AML) screening checking whether any suspect wallet is listed on US OFAC SDN, UN Sanctions, EU Blacklists, or FIU-IND registries.

Real-World Example: Scanning a ransomware address instantly displays a red badge: '**OFAC SDN Sanctioned — DPRK State Sponsored Threat**', alerting police to involve international agencies.

■ 14. Live Mempool Monitoring Stream

What it does: Live blockchain feed that watches flagged wallets in real-time and alerts investigators with visual pulse beacons the exact second new transactions are broadcast.

Real-World Example: While an officer is investigating Case #1245, a new ■80,000 transaction is broadcast to the mempool. The live feed lights up green: 'Live Event: ■80,000 moved to Binance Hot 14', enabling instant action.

■ 15. Investigator Case Workspace & Evidence Vault

What it does: Multi-case organizer where police officers can manage multiple FIRs, assign junior officers, write encrypted case notes, and maintain an audit log of evidence.

Real-World Example: Officer records notes: 'Requested Binance KYC for UID #8912401 on 24 Aug. Bank account frozen via 1930 portal.' Notes are encrypted and stored in the local vault.

■ 16. Multi-Jurisdictional Legal Subpoena Dispatcher

What it does: 1-click generator and emergency API dispatcher that formats complete legal notices under Section 91 CrPC (India), US CLOUD Act § 2703(d), and MLAT freeze orders.

Real-World Example: Automatically generates a formal Section 91 CrPC notice pre-filled with the transaction hash, deposit amount (■20,000), timestamp, and sends it directly to Binance's Nodal Officer.

■ 17. Public Wallet Safety Check ('Check Before You Send')

What it does: A public consumer protection tool where citizens can paste any wallet address before paying someone online to verify if it is safe or a reported scam.

Real-World Example: A student is asked to send ■10,000 to a crypto wallet for a 'Telegram job'. They check the address on CyberTrace, which displays: '■ DO NOT SEND — 91% Match with Known Fraud Syndicate', preventing the loss before it happens.

■ 18. Real-Time On-Chain RPC Querying (Live Web3 Mode)

What it does: Directly queries live Ethereum and BSC mainnet RPCs in real-time, fetching real on-chain ETH balances, nonces, and transaction counts for any live wallet.

Real-World Example: Entering Vitalik Buterin's address (0xd8dA...6045) queries Ethereum mainnet and displays real live balance: **6.6410 ETH (■18,26,275)** across 5,956 live transactions.

5. Enterprise Security Layer & Anti-Hacker Defense

Multi-Layered Client-Side & Network Defense:

- **Anti-XSS Sanitizer:** Strict HTML entity encoding & regex stripping prevents script injection across all search and reporting fields.
- **Anti-Clickjacking Framebuster:** X-Frame-Options: DENY and runtime JavaScript framebuster prevent unauthorized iframe embedding.
- **Content Security Policy (CSP Level 3):** Restricts network sockets strictly to whitelisted verified Ethereum/BSC JSON-RPC nodes.
- **Token Bucket Rate Limiter:** Throttles automated bot floods and brute-force scanning (50 req/min max).
- **AES-256 / SHA-256 Cryptographic Vault:** Case notes, FIR logs, and Section 91 CrPC evidence dossiers are cryptographically sealed.

6. Quick Reference Links & Verification Resources

Live Platform (GitHub Pages):	https://hidayatulla268.github.io/CyberTrace/
GitHub Repository:	https://github.com/Hidayatulla268/CyberTrace
Documentation PDF File:	CyberTrace Platform Guide.pdf
Security Certification:	Enterprise Grade (Anti-XSS • CSP Level 3 • SHA-256 Vault)

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